

Differential Topology 1

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Chapter 1

Intro

1.1 Topological manifolds

Lecture 1: Syllabus day

Wed 03 Sep 2025 12:16

Smooth manifolds form a proper subset of topological manifolds. Interesting.

Definition 1.1.1. A topological manifold is a Hausdorff topological space $(M, \mathcal{T})$ which is second countable (it has a countable basis), and for some fixed $n \in \mathbb{N}$, for all points $p \in M$ there is an open set $U \subseteq M$ and a homeomorphism $f : U \cong \hat{U} \subseteq \mathbb{R}^n$ for some $\hat{U}$. We say the dimension of M is n , and call it an n -manifold.

Of course we may equivalently say for all points p , there is an open neighborhood U and a homeomorphism $f : U \cong \mathbb{R}^n$. This follows since $\hat{U}$ admits an open cover by open balls, which have open preimages, and thus at least one has an open preimage V containing p . Since the image of $f|_V$ is an open ball, and open balls are homeomorphic to $\mathbb{R}^n$, the conclusion follows.

Lecture 2: Smooth manifolds

Fri 05 Sep 2025 12:20

Given $f : M \rightarrow \mathbb{R}$ and a point p , we might want to differentiate f at p . We can take an open neighborhood $p \in U$ and a homeomorphism $\varphi : U \cong \tilde{U} \subseteq \mathbb{R}^n$, and try to differentiate $f \circ \varphi^{-1}$. The issue is that given another open set V and another homeomorphism ψ , the change of variables from $f \circ \varphi^{-1}$ to $f \circ \psi^{-1}$ might not be smooth.

Lecture 3: Smooth Manifolds

Mon 08 Sep 2025 12:22

we want to define what it means for a function $f : M \rightarrow \mathbb{R}$ to be smooth (infinitely differentiable) at some $p \in M$. Since M is a manifold, there is an open set $p \in U \subseteq M$ and a homeomorphism $\varphi : U \cong \tilde{U} \subseteq \mathbb{R}^n$, so we can consider the function $f \circ \varphi^{-1} : \tilde{U} \rightarrow \mathbb{R}$. We can compute the partials of this function.

Terminology 1.1.2. We say f is *smooth* or C^∞ at $p \in U$ if $f \circ \varphi^{-1}$ is C^∞ at $\hat{p} = \varphi(p) \in \tilde{U}$.

The problem is that although $f \circ \varphi^{-1}$ might be smooth at $\hat{p}$, but given another chart $\psi : V \rightarrow \tilde{V}$, $f \circ \psi^{-1}$ might not be smooth at $\psi(p)$. We need to add more structure to M for this notion of smoothness to be well-defined.

Note that we can restrict $\psi \circ \varphi^{-1}$ to $\psi \circ \varphi^{-1} : \varphi(U \cap V) \rightarrow \psi(U \cap V)$. Of course $f \circ \psi^{-1} = f \circ \varphi^{-1} \circ \varphi \circ \psi^{-1}$. The issue can be rephrased: although $f \circ \varphi^{-1}$ might be C^∞ , $\varphi \circ \psi^{-1}$ might not be. The component $\varphi \circ \psi^{-1}$ is the *change of variables*, and we need the change of variables to be smooth. To ensure this is true, we *choose* a collection of homeomorphisms that are smooth, and consider that structure on the manifold, instead of taking *all* local homeomorphisms.

Definition 1.1.3. A *chart* for a point $p \in M$ is an open set $p \in U \subseteq M$ and a homeomorphism $\varphi : U \cong \tilde{U} \subseteq \mathbb{R}^n$.

Definition 1.1.4. Let $U, V \subseteq M$ be open and $\varphi : U \rightarrow \varphi(U) \subseteq \mathbb{R}^n$, $\psi : V \rightarrow \psi(V) \subseteq \mathbb{R}^n$ be charts. We say φ and ψ are C^∞ -related if $\varphi \circ \psi^{-1}$ and $\psi \circ \varphi^{-1}$ are both C^∞ in the sense that all of their partials exist.

Definition 1.1.5. Let M be a topological manifold. An *atlas* $\mathcal{A}$ (sometimes a *smooth atlas*) is a collection $\{U_\alpha, \varphi_\alpha\}_{\alpha \in A}$ of charts such that

1. The charts form an open cover of M :

$$M = \bigcup_{\alpha \in A} U_\alpha;$$

2. Any pair of charts is C^∞ .

Terminology 1.1.6. We often call each chart U_α, φ_α a *coordinate system*.

We now know what it means for a map $f : M \rightarrow \mathbb{R}$ to be smooth: we just require it be smooth with respect to an atlas $\mathcal{A}$ for M . However, smooth manifolds has even more structure. One can choose *larger* atlases.

Definition 1.1.7. Given an atlas $\mathcal{A}$ for a manifold M , define the *maximal atlas* $\mathcal{A}'$ containing $\mathcal{A}$ to be the collection of *all* charts on M which are C^∞ -related to *all* charts in $\mathcal{A}$.

This condition guarentees we always have enough charts for our atlas M . This leads us to

Definition 1.1.8 (Smooth manifold). A *smooth n -manifold* (also a C^∞ n -manifold) is a pair $(M, \mathcal{A})$ where M is an n -manifold and $\mathcal{A}$ is a maximal atlas on M .

Example 1.1.9. A 0-manifold is the *same* as a smooth 0-manifold and contains a countable discrete set.

Consider $1 : \mathbb{R} \rightarrow \mathbb{R}$, and consider another homeomorphism $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ given by $x \mapsto x^3$. These are both certainly charts, and $\{(\mathbb{R}, \psi)\}$ certainly yields a smooth atlas. This is interestingly a maximal atlas?????

Consider matrices $M_{m \times n}(\mathbb{R}) \cong \mathbb{R}^{mn}$. This is also a smooth manifold.

Consider the graph of a C^∞ function $f : U \subseteq \mathbb{R}^k \rightarrow \mathbb{R}^n$. For simplicity, write Γ_f for the graph of f . We claim that this is a smooth manifold of dimension k . The natural chart is given by composing along the natural projection $\pi : \Gamma_f \rightarrow \mathbb{R}^k$. Collecting these gives an atlas.

Example 1.1.10. An important example is the standard smooth structure on S^n . Recall that we have for any $0 < j \leq n+1$, a homeomorphism $\varphi_j^\pm : U_j^\pm \rightarrow \tilde{U}_j^\pm$ given by taking the upper hemisphere (+) or lower hemisphere (−) and mapping them down to a unit disk. One rigorously defines this as the product of 1_i for $i \neq j$, and restricting along the relevant (open) hemisphere. Alternatively, $\varphi_j^\pm : (x^1, \dots, x^n) \mapsto (x^1, \dots, x^{j-1}, x^{j+1}, \dots, x^n)$.

We claim the collection $\{\varphi_j^\pm\}_{j \in [n+1] \setminus [0]}$ is a smooth atlas. We check this in cases. Take $i < j$, assume WLOG that both are in the upper hemisphere, and consider $\varphi_i \circ \varphi_j^{-1}$. We show these are both smooth. To construct φ_j^{-1} , we need to re-insert the component making the norm of each vector 1. We thus have

$$\varphi_i \circ \varphi_j : (x^1, \dots, x^{j-1}, x^{j+1}, \dots, x^n) \mapsto (x^1, \dots, x^{i-1}, x^{i+1}, \dots, x^{j-1}, 1 - \sqrt{\sum_{k=1}^n (x^k)^2}, x^{j+1}, \dots, x^n).$$

This is smooth since it is a product of smooth functions. Thus, φ_j is C^∞ -related to φ_i .

This argument immediately generalizes to *level sets*. Given $C^\infty \Phi : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ with preimage $M_c := \Phi^{-1}(c)$ as a *level set* (norm is 1) and nonzero derivatives at all $a \in M_c$, we have by the implicit function theorem that M is a C^∞ $(n-1)$ -manifold.

Lecture 4: awoooo

Wed 10 Sep 2025 12:11

Let's now show $\mathbb{RP}^n$ is a topological manifold. Recall the real projective space is the quotient of $\mathbb{R}^{n+1}$ by identifying colinear n -tuples (in the linear algebraic sense of "colinear", meaning the line passes through 0). Choose $U_\alpha \subseteq \mathbb{RP}^n$, and define a chart $\varphi_\alpha : U \rightarrow \mathbb{R}^n$ by scalar multiplying by $1/z^\alpha$ for z^α the α th component of a choice of representative for a point in U_α . This is a smooth homeomorphism apparently.

The product $M \times N$ of smooth manifolds is smooth.

Let's specialize temporarily to manifolds with boundaries. We start with topological manifolds. We write ∂M for the boundary of M . Write $\mathbb{H}^n$ for the upper half space $x \in \mathbb{R}^n, x^n \geq 0$.

Definition 1.1.11. A topological manifold with boundary is a second countable, Hausdorff space M wherein every point of M has a neighborhood homeomorphic to an open set of $\mathbb{H}^n$.

Manifolds with boundary then have two different types of charts: interior and boundary charts, defined in the obvious way. An *interior* point lives in some interior chart, otherwise it is a boundary point. We write $\text{Int}(M)$ for the set of interior points. We need to know what it means to be smooth on a boundary chart. The solution is simply saying it is smooth at a point *if and only if* it admits an extension to an open neighborhood of a point.

1.2 Smooth maps

Definition 1.2.1. Let M, N be m - and n -manifolds, respectively. A map $f : M \rightarrow N$ is *smooth* at some $p \in M$ if and only if given a chart (U, φ) of p in the atlas, and some chart (V, ψ) of $f(p)$ in the atlas of N , the composite

$$\mathbb{R}^n \supseteq \tilde{U} \xrightarrow{\varphi^{-1}} M \xrightarrow{f} N \xrightarrow{\psi} \tilde{V} \subseteq \mathbb{R}^m$$

is smooth in the usual sense, *and* if $f(U) \subseteq V$. If f is smooth at all $p \in M$, we say f is a smooth map.

Definition 1.2.2. Let M be a smooth manifold. A smooth map $f : M \rightarrow \mathbb{R}$ will be called a *smooth function* on M . The set of all smooth functions on M is written $C^\infty(M)$.

$C^\infty(M)$ forms an $\mathbb{R}$ -algebra in the usual way.

Definition 1.2.3. Let $f : M \rightarrow N$ be smooth. Then f is a *diffeomorphism* if it is invertible and its inverse is smooth. If there is a diffeomorphism $M \rightarrow N$, we say M and N are *diffeomorphic*, and write $M \cong N$.

Theorem 1.2.4. *Composites of smooth maps are smooth.*

Proof. Let $f : M \rightarrow N$ and $g : N \rightarrow R$ be smooth maps. Let $p \in M$, and since f is smooth, there is a chart (U, φ) for p and a chart (V, ψ) of $f(p)$ with $\psi \circ f \circ \varphi^{-1}$ smooth, and $f(U) \subseteq V$. Since g is smooth, there is a chart (V', ψ') of $f(p)$ and a chart (W, ξ) of $gf(p)$ such that $g(V') \subseteq W$ and such that $\xi \circ g \circ (\psi')^{-1}$ is smooth. If we show that (V, ψ) suffices as a choice of (V', ψ) , then we have $gf(U) \subseteq g(V) \subseteq W$ and $\xi \circ g \circ \psi^{-1} \circ \psi \circ f \circ \varphi^{-1} = \xi \circ gf \circ \varphi^{-1}$ is smooth. Thus, it suffices to show that statement.

use the transitio map ■

Remark 1.2.5. doesnt matter what chart we choose cuz theyre all smoothly related!!! rigorize this

Remark 1.2.6. inclusions and projections (in the sense of subobjects and quotients) are differentiable

Lecture 5: Tangent vectors

Fri 12 Sep 2025 12:21

Define $\mathbb{R}_a^n$ as the set $\{a\} \times \mathbb{R}^n$ for some vector $a \in \mathbb{R}^n$. We will show that this admits an interpretation as a tangent space to $\mathbb{R}^n$ at a . We say that an element $(a, v) \in \mathbb{R}_a^n$ is a *geometric tangent vector*, and we write it as v_a or $v|_a$. The intuitive interpretation of this object is taking the vector v , and giving it starting point a .

The set $\mathbb{R}_a^n$ forms a vector space in the obvious way, by interpreting $\{a\}$ as 0, and simply working with the space $0 \times \mathbb{R}^n \cong \mathbb{R}^n$. The vectors $e_i|_a$ thus form a basis for this space.

This allows us to define tangent spaces to manifolds which admit embeddings into $\mathbb{R}^{n+1}$, by taking the tangent space at a point $a \in M$ as $\mathbb{R}_a^n$. However, to generalize this definition arbitrarily, we must explore some alternative representations of this information.

To do this, we must somehow translate this information into that of smooth maps and manifolds. For example, given any vector $v_a \in \mathbb{R}_a^n$, the directional derivative $D_v|_a : C^\infty(\mathbb{R}^n) \rightarrow \mathbb{R}$ given by

$$D_v|_a f = D_v f(a) = \left. \frac{d}{dt} \right|_{t=0} f(a + tv)$$

is $\mathbb{R}$ -linear, and satisfies the *product rule*

$$D_v|_a (fg) = f(a)D_v|_a g + g(a)D_v|_a f.$$

If we write $v_a = v^i e_i|_a$, then $D_v|_a f$ can be written more concretely as

$$D_v|_a f = v^i \frac{\partial f}{\partial x^i}(a).$$

This follows by the chain rule. We generalize directional derivatives using this formulation.

Definition 1.2.7. Let $a \in \mathbb{R}^n$. A *derivation* at a is an $\mathbb{R}$ -linear map $w : C^\infty(\mathbb{R}^n) \rightarrow \mathbb{R}$ satisfying the product rule

$$w(fg) = f(a)w(g) + g(a)w(f).$$

We write $T_a \mathbb{R}^n$ for the set of all derivations at a .

We immediately see that $T_a\mathbb{R}^n$ is a vector space by noting that if w_1, w_2 are derivations, then

$$(w_1+w_2)(fg) = (f(a)w_1(g)+g(a)w_1(f))+(f(a)w_2(g)+g(a)w_2(f)) = f(a)(w_1+w_2)(g)+g(a)(w_1+w_2)(f),$$

$$(cw)(fg) = c(f(a)w(g) + g(a)w(f)) = f(a)(cw)(g) + g(a)(cw)(f).$$

The following lemma is not too interesting, but implies a very interesting result.

Lemma 1.2.8. *Let $a \in \mathbb{R}^n$, $w \in T_a\mathbb{R}^n$, and $f, g \in C^\infty(\mathbb{R}^n)$. If f is a constant function, then $w(f) = 0$. Additionally, if $f(a) = g(a) = 0$, then $w(fg) = 0$.*

Proof. For the first statement, it suffices to prove f for the constant function $f(x) = 1$, since $g(x) = c$ implies $g = cf$, and the statement follows by linearity of derivations. The product rule gives

$$w(f) = w(ff) = 1w(f) + 1w(f) = 2w(f)$$

and thus $w(f) = 0$.

For the second statement, we have

$$w(fg) = f(a)w(g) + g(a)w(f) = 0w(f) + 0w(g) = 0.$$

■

Proposition 1.2.9. *Let $a \in \mathbb{R}^n$ and $v_a \in \mathbb{R}_a^n$. Then $D_v|_a$ is a derivation, and the map $v_a \mapsto D_v|_a$ is an isomorphism of $\mathbb{R}$ -vector spaces $\mathbb{R}_a^n \cong T_a\mathbb{R}^n$.*

Proof. First, we show $D_v|_a$ is a derivation. This follows by the properties of the partial derivative ∂_μ . To show it is $\mathbb{R}$ -linear, we have

$$D_v|_a(f + g) = v^\mu \partial_\mu (f + g)|_a = v^\mu \partial_\mu f|_a + v^\mu \partial_\mu g|_a = D_v|_a f + D_v|_a g,$$

$$D_v(cf) = v^\mu \partial_\mu (cf)|_a = cv^\mu \partial_\mu f|_a = cD_v|_a f.$$

The product rule also follows by the fact that ∂_μ satisfies the product rule: $\partial_x(fg) = f\partial_x(g) + g\partial_x f$. Adding in the evaluation gives

$$D_v|_a(fg) = v^\mu (f(a)\partial_\mu g|_a + g(a)\partial_\mu f|_a) = f(a)D_v|_a g + g(a)D_v|_a f.$$

Now we must show the map $v_a \mapsto D_v|_a$ is $\mathbb{R}$ -linear. To show additivity, we must show $D_{v+w}|_a = D_v|_a + D_w|_a$. This is obvious by the fact that $(v + w)^\mu = v^\mu + w^\mu$:

$$D_{v+w}|_a = (v + w)^\mu \partial_\mu|_a = v^\mu \partial_\mu|_a + w^\mu \partial_\mu|_a.$$

Similarly, since $(cv)^\mu = cv^\mu$, we have

$$D_{cv}|_a = (cv)^\mu \partial_\mu|_a = cv^\mu \partial_\mu|_a = cD_v|_a.$$

Therefore, the map is $\mathbb{R}$ -linear.

The main part of the proof is that the map is an isomorphism of $\mathbb{R}$ -vector spaces. Injectivity is shown by showing the kernel is trivial. Since we do not know that $T_a\mathbb{R}^n$ is finite dimensional, this does not suffice as an isomorphism proof. Let $D_v|_a$ be the zero derivation $f \mapsto 0$. We want to show $v_a = 0$. Let f be the coordinate function $x^j : \mathbb{R}^n \rightarrow \mathbb{R}$ by projecting onto the j th component. Then $\partial_i x^j = \delta_i^j$, giving

$$0 = D_v|_a = v^\mu \partial_\mu x^j|_a = v^j,$$

since $1|_a = 1$. The statement follows by letting j range over $[n]$, showing the map is injective.

To prove surjectivity, let $w \in T_a \mathbb{R}^n$ be an arbitrary derivation. With the same definition of x^j , write $v^i = w(x^i)$. We show $w = D_v|_a$.

We show this by proving they act on maps $f : C^\infty(\mathbb{R}^n)$ in the same way. Since f is analytic, it admits a Taylor expansion

$$f = f(a) + \sum_{i=1}^n \frac{\partial f}{\partial x^i}(a) (x^i - a^i) + \sum_{i,j=1}^n (x^i - a^i) (x^j - a^j) \int_0^1 (1-t) \frac{\partial^2 f}{\partial x^i \partial x^j}(a + t(x-a)) dt.$$

The functions $x^i - a^i$ and $x^j - a^j$ vanish at $x = a$. Since w is $\mathbb{R}$ -linear and distributes over addition, by lemma 1.2.8,

$$w((x^i - a^i)(x^j - a^j)) = 0.$$

The derivation thus annihilates the last term:

$$w(f) = w(f)(a) + \sum_{i=1}^n w\left(\frac{\partial f}{\partial x^i}(a) (x^i - a^i)\right) = \sum_{i=1}^n \frac{\partial f}{\partial x^i}(a) (w(x^i) - w(a^i)) = \sum_{i=1}^n \frac{\partial f}{\partial x^i}(a) w(x^i).$$

Since $w(x^i) = v^i$, this is simply $v^\mu \partial_\mu f|_a = D_v|_a$. This proves surjectivity. ■

Corollary 1.2.10. $T_a \mathbb{R}^n$ has dimension n , and has a basis given by

$$\left. \frac{\partial}{\partial x^i} \right|_a := D_{e_i}|_a$$

In particular, these are defined by

$$\left. \frac{\partial}{\partial x^i} \right|_a f = \frac{\partial f}{\partial x^i}(a).$$

Proof. These follow immediately by the above proof. ■

We now see an obvious way to generalize this definition to arbitrary manifolds. We can define derivations on arbitrary C^∞ -manifolds, and then show that this is generated by the partials at a point.

Definition 1.2.11. Let M be a smooth manifold, and let $p \in M$. A linear map $v : C^\infty(M) \rightarrow \mathbb{R}$ is a *derivation at p* if it satisfies the product rule

$$\forall f, g \in C^\infty(M) : v(fg) = f(p)v(g) = g(p)v(f).$$

The set of all derivations for p is denoted by $T_p M$, and is called the *tangent space* to M at p . An element of $T_p M$ is a *tangent vector* at p .

We now prove some facts to show that this definition behaves as expected.

Lemma 1.2.12. Let M be a smooth manifold with or without boundary, $p \in M$, $v \in T_p M$, and $f, g \in C^\infty(M)$. If f is constant, then $v(f) = 0$. If $f(p) = g(p) = 0$, then $v(fg) = 0$.

Proof. The same proof for lemma 1.2.8 suffices. ■

Lecture 6: Tangent space

Mon 15 Sep 2025 12:22

But how do we express some $v \in T_p M$ in terms of the coordinates x^i of a given chart φ ($\varphi_i = x^i$)? Given the diagram

$$\begin{array}{ccc} U & \xrightarrow{\varphi} & \tilde{U} \\ \downarrow & & \downarrow \\ M & & \mathbb{R}^n \end{array}$$

the tangent space induces a diagram

$$\begin{array}{ccc} T_p U & \xrightarrow{\varphi_*} & T_p \tilde{U} \\ \downarrow i_* & & \downarrow i_* \\ T_p M & & T_p \mathbb{R}^n \end{array}$$

We would like to show there is an isomorphism $T_p M \cong T_p \mathbb{R}^n$, rendering the diagram commutative.

Definition 1.2.13. Let M be a space and $f : M \rightarrow \mathbb{R}^n$ (often $\mathbb{R}$). The *support* of f , denoted $\text{supp } f$, is the closure of the set of points where f is nonzero:

$$\text{supp } f = \overline{\{p \in M \mid f(p) \neq 0\}}.$$

If $\text{supp } f \subseteq U$ for some open U , we say f is *supported in* U .

Definition 1.2.14. Let M be a smooth manifold, and $A \subseteq M$ a closed subset contained in some open set U . A continuous function $\psi : M \rightarrow \mathbb{R}$ is a *bump function* for A supported in U if and only if $0 \leq \psi \leq 1$, and $\psi|_A = 1$ with $\text{supp } \psi \subseteq U$.

Proposition 1.2.15. Let $p \in M \in \text{Man}$ and $v \in T_p M$. If $f, g \in C^\infty(M)$ agree on some neighborhood of p , then $v(f) = v(g)$. We say that $T_p M$ depends only on a neighborhood of p .

Proof. Let $f|_U = g|_U$ with $p \in U$ open in M . Then defining $h = f - g$, we see that $h|_U = 0$. Let $\text{supp } h$ be some closed subset containing U . We claim there exists a smooth bump function for $\text{supp } h$ supported on U . This follows by prop 2.2.5 (READ).

Let $\psi \in C^\infty(M)$ be a smooth bump function for $A = \text{supp } h$ and is supported in U . Then $\psi h = h$ since $\psi = 1$ where h is nonzero. Then $v(\psi h) = 0$ by product rule. ????????/ ■

Proposition 1.2.16. Let M be a smooth manifold, $U \subseteq M$ an open submanifold, $\iota : U \hookrightarrow M$ the natural inclusion, and $p \in U$. Then $\iota_* : T_p U \rightarrow T_p M$ is an isomorphism of vector spaces.

Proof. Injectivity: The key is the *extension lemma*. Let $p \in V \subseteq U$ such that $\bar{V} \subseteq U$. Then Jesus christ i am just going to read the book. ■

Corollary 1.2.17. If M is a smooth n -manifold, then $\dim_{\mathbb{R}} T_p M = n$ for any $p \in M$.

This corollary gives a basis for the space. Let $\varphi : T_p M \cong T_p \mathbb{R}^n$. Since the latter has as a basis the set $\partial_\mu|_{\hat{p}}$, meaning we have a basis in $T_p M$ by

$$\varphi_*^{-1} \left(\frac{\partial}{\partial x^i} \Big|_{\varphi(p)} \right).$$

This is a lot of notation, so we simply write

$$\left. \frac{\partial}{\partial x^i} \right|_p \quad \text{or} \quad \partial_i|_p.$$

A vector $v \in T_p M$ is then written as

$$V = V^i \partial_i|_p = V^i \varphi_*^{-1} \left(\left. \frac{\partial}{\partial x^i} \right|_p \right).$$

Note that V^i are literal real numbers since they are just coefficients for the basis.

Now consider a manifold M with boundary. The definition of $T_p M$ still works. Since the space has boundary, it is locally homeomorphic to an open set in the half space $\mathbb{H}^n$. This includes into $\mathbb{R}^n$, and since smooth functions on the half space must extend to smooth maps on $\mathbb{R}^n$, we still get our definition.

Here's a useful special case. Let V a vector space and $a \in V$. There exists a natural isomorphism of vector spaces $V \rightarrow T_a V$ given by $a \mapsto D_v|_a$ natural in both V and a . This map is natural $1_{\mathbb{R}\text{-Vect}} \cong T_{(-)}(-)$.

Does the tangent space functor preserve products?

Proposition 1.2.18. *T preserves products (3.14).*

Proof. I think what we do is bring it into $\mathbb{R}^n$, distribute there, and pull back.23212 ■

Lecture 7: Tangent bundles

Fri 19 Sep 2025 12:20

I MISSED STUFF. WE PICKED UP WITH CHANGE OF COORDINATES IN TANGENT SPACES $T_p M$.

Choose $p \in M$, and choose a chart $(U, \varphi = u^\mu)$ for p . This is a coordinate system, and $\varphi(p) = \hat{p} = (x^1, \dots, x^n)$ in the natural basis $\partial_\mu|_{\hat{p}}$. This gives a vector

$$\left. \frac{\partial}{\partial x^i} \right|_p = \varphi^{-1} \left(\left. \frac{\partial}{\partial x^i} \right|_{\hat{p}} \right) \implies v = v^\mu \partial_\mu.$$

However, given any other chart (V, ψ) , we get another basis induced in the same way. To change variables, we use the precomposition-esek change of variables map $(\psi \circ \varphi^{-1})_*$, which is literally the change of variables in the calc 3 sense.

This change of variables basically becomes the chain rule.

Lemma 1.2.19. *Every $v \in T_p M$ is tangent to some smooth curve in M .*

Proof. Let $(p \in U, \varphi)$ a chart of the curve at p . Then

$$v = v^i \left. \frac{\partial}{\partial x^i} \right|_p.$$

Let $\gamma : (-\varepsilon, \varepsilon) \rightarrow M$ have $\gamma_i(t) = t v^i$. Then $\gamma(0) = p$ and γ is C^∞ . Add idk i think we are writing γ' instead which is $\gamma_* \frac{d}{dt}$ eval at p ? ■

Definition 1.2.20. Let M be a smooth manifold. Define the *tangent bundle* TM as the disjoint union

$$TM := \coprod_{p \in M} T_p M.$$

The motivation here is smooth vector fields. There is a projection $TM \rightarrow M$ by mapping each $v \in T_p M \subseteq TM$ to p :

$$\pi : T_p M \mapsto p.$$

If we give TM a smooth structure, we can define some $V : M \rightarrow TM$ to be a smooth vector field if $\pi \circ V = 1$.

Proposition 1.2.21. *For any smooth n -manifold M , the tangent bundle has a natural smooth structure making it a $2n$ -manifold, and the canonical projection $\pi : TM \rightarrow M$ is smooth.*

Lecture 8: Tangent bundles

Mon 22 Sep 2025 12:20

We want to give TM a smooth manifold structure such that proposition 1.2.21 is true. Our idea is if (U, φ) is a chart, consider $\pi^{-1}(U)$, which is the set of tangent vectors at $p \in U$ for all p . We can define a function

$$\tilde{\varphi} : \pi^{-1}(U) \rightarrow \tilde{U} \times \mathbb{R}^n \subseteq \mathbb{R}^{2n}, \quad T_p M \ni v \mapsto (\varphi(p), v^1, \dots, v^n),$$

where v^i is given under the chart (U, φ) . We want to topologize TM using the maps $\tilde{\varphi}$. To do this, we demand that $\tilde{\varphi}$ be a homeomorphism. We recall a lemma from earlier in the book.

Lemma 1.2.22. *Let M be a set, and $\{U_\alpha\}_{\alpha \in A}$ a collection of subsets of M , together with injective maps $\varphi_\alpha : U_\alpha \rightarrow \mathbb{R}^n$ for all $\alpha \in A$, such that the following holds.*

1. *For all $\alpha \in A$, $\varphi_\alpha(U_\alpha)$ is open.*
2. *For all $\alpha, \beta \in A$, $\varphi_\alpha(U_\alpha \cap U_\beta)$ and $\varphi_\beta(U_\alpha \cap U_\beta)$ is open.*
3. *Whenever the intersection is nonempty, $\varphi_\alpha \circ \varphi_\beta^{-1}$ is a diffeomorphism.*
4. *Countably many U_α s cover M .*
5. *For all $p \neq q \in M$, either there is some $U_\alpha \ni p, q$, or there are disjoint sets U_α, U_β with $p \in U_\alpha$ and $q \in U_\beta$.*

Then M has a unique smooth structure such that each $(U_\alpha, \varphi_\alpha)$ is a smooth chart.

Proof. Topologize by generating a topology on preimages of each φ_α on open sets. Second countability follows by (4). Hausdorff follows by (5) and the definition of open sets. Since each φ_α is injective, factoring through the image gives $\varphi_\alpha : U_\alpha \rightarrow \varphi_\alpha(U_\alpha)$ is a homeomorphism. We then obtain the smooth transition maps by (2) and (3). ■

Recall our situation with TM . For each chart (U, φ) of M , there is an injective map $\tilde{\varphi} : \pi^{-1}(U) \rightarrow \tilde{U} \times \mathbb{R}^n$. Then the collection of $(\tilde{\varphi}, \pi^{-1}(U))$ satisfies the requirements of lemma 1.2.22. Smoothness of π is immediate. Additionally,

Proposition 1.2.23. *If M is a smooth n -manifold which can be covered by one chart, then there is a diffeomorphism $TM \cong M \times \mathbb{R}^n$.*

Proof. Follows by definition, since there is a chart $\tilde{\varphi} : TM \rightarrow M \times \mathbb{R}^n$ since the chart covers the manifold. The global chart is a diffeomorphism. We then say that TM is a trivial bundle. ■

The upshot of this is for all $p \in M$, there is a neighborhood U of p with $\pi^{-1}(U) \cong U \times \mathbb{R}^n$. Thus, TM is locally trivial.

This puts constraints on what TM can be. Consider TS^1 . If we imagine gluing a line to each point of S^1 perpendicular to the plane of the circle, we get a cylinder. This is correct, since $TS^1 \cong S^1 \times \mathbb{R}$. However, we could also glue them by rotating the direction as we traverse around the circle, giving us a Möbius strip, which we cannot have since it is not locally trivial.

Definition 1.2.24. Let $F : M \rightarrow N$ be a smooth map. Define $F_* = dF$, called the *global pushforward* or *global differential* by $dF : TM \rightarrow TN$ via $dF|_{T_p M} = dF_p$. Alternatively,

$$dF = \coprod_{p \in M} dF_p.$$

This still behaves covariantly.

Definition 1.2.25. A smooth vector field V is a smooth map $M \rightarrow TM$ such that $\pi \circ V = 1$.

Definition 1.2.26. Let $F : M \rightarrow N$ be a smooth map. Then the *rank* of F at p is the rank of the matrix representation of $dF_p : T_p M \rightarrow T_p N$. If the rank is constant with respect to p , we say the rank is constant, and simply refer to the *rank* of F . The rank at p is written $\text{rk}_p F$.

Clearly, given $F : M \rightarrow N$,

$$0 \leq \text{rk}_p F \leq \min(\dim M, \dim N).$$

If $\text{rk}_p F = \min(\dim M, \dim N)$, then we say F has *full rank* at p .

Definition 1.2.27. A smooth map $F : M \rightarrow N$ is a *submersion* if and only if F is constant rank, and $\text{rk} F = \dim N$. We say F is a *immersion* if and only if F is constant rank, and $\text{rk} F = \dim M$.

Proposition 1.2.28. Let $F : M \rightarrow N$ be smooth, and let $p \in M$. If $dF_p : T_p M \rightarrow T_{F(p)} N$ is surjective, then there is a neighborhood U of p such that $F|_U$ is a submersion. If it is injective, then $F|_U$ is an immersion.

Proof. Obvious, since the rank of surjections is just $\dim$ of codomain, similarly for injections, and continuity of F implies there is a neighborhood wherein rank is constant. And also the fact that the space of full rank matrices is open. ■

For example, the projection $\pi : M \times N \rightarrow M$ is surjective, so it is surjective on all tangent spaces. Thus, π is a smooth submersion. An immersion could be given by a smooth path $\gamma : I \rightarrow M$ if and only if it has everywhere nonzero velocity, since $d\gamma$ is non-rank-0 whenever velocity doesn't vanish everywhere. A fun example is $\pi : TM \rightarrow M$ is a smooth submersion. This follows since it's a local property, so we apply the definition.

Lecture 9: Submersions and immersions

Wed 24 Sep 2025 12:24

The linear map dF as a matrix in a basis is the Jacobian

$$J_{ij} = \frac{\partial \hat{F}^i}{\partial x^j}.$$

Definition 1.2.29. Let $F : M \rightarrow N$ be smooth. Then F is a *local diffeomorphism* if and only if for all $p \in M$, there is some $U \subseteq M$ open such that $F(U)$ is open in N , and $F|_U$ is a diffeomorphism onto its image.

For example, consider the covering $\mathbb{R} \rightarrow S^1$. This is clearly not a diffeomorphism, but it is a local diffeomorphism since we just restrict to a small enough neighborhood such that it becomes injective.

Theorem 1.2.30. Let M, N be smooth manifolds, $F : M \rightarrow N$ smooth, and $p \in M$ such that dF_p is invertible. Then there is a connected neighborhood U_0 of p , and V_0 of $F(p)$, such that $F|_{U_0} : U_0 \cong V_0$ is a diffeomorphism.

Here are some obvious properties of local diffeomorphisms.

- Composites of local diffeomorphisms are locally diffeomorphic.
- Finite products of local diffeomorphisms are locally diffeomorphic.
- Diffeomorphisms are local diffeomorphisms.
- Every bijective local diffeomorphism is a diffeomorphism.

Proposition 1.2.31. Let $F : M \rightarrow N$ be smooth. Then F is a local diffeomorphism if and only if F is an immersion and a submersion. Moreover, if $\dim M = \dim N$ and F is either an immersion or a submersion, then it is a local diffeomorphism.

Theorem 1.2.32 (Rank). Let $F : M^m \rightarrow N^n$ be smooth and constant rank $\text{rk } F = r$. Then for all $p \in M$, there are two smooth charts (U, φ) centered at p (meaning $\varphi(p) = 0$), and (V, ψ) or N centered at $F(p)$, such that $(\psi \circ F \circ \varphi^{-1})(x^1, \dots, x^m) = (x^1, \dots, x^r, 0, \dots, 0)$.

In particular, if F is a submersion ($n \leq m$), then $\hat{F}(x^1, \dots, x^m) = (x^1, \dots, x^n)$. If F is an immersion ($m \leq n$), then $\hat{F}(x^1, \dots, x^m) = (x^1, \dots, x^m)$. Smooth submersions thus look like projections, and smooth immersions look like injections. Constant rank maps are thus in particular composites of submersions and immersions.

Proof. WLOG, take $M = U \subseteq \mathbb{R}^n$ open, $V \subseteq \mathbb{R}^n$ open, and let $F : U \rightarrow V$ be smooth. ■

Lecture 10: The rank theorem

Fri 26 Sep 2025 12:21

The rank theorem tells us that immersions are related to embeddings, and submersions are related to quotients. Indeed, we *define* smooth embeddings as embeddings which are also immersions. Of course, these are just injective immersions.

Lecture 11: ??

Mon 29 Sep 2025 12:36

Theorem 1.2.33. Let $F : M \rightarrow N$ smooth. Then F is a smooth immersion if and only if every $p \in M$ has a neighborhood $U \subseteq M$ such that $F|_U$ is a smooth embedding.

Theorem 1.2.34 (Local section theorem). Let $\pi : M \rightarrow N$ be smooth. Then π is a smooth submersion if and only if every $p \in M$ is in the image of a smooth local section of π .

Smooth submersions are open, and if they are surjective then they are quotients.

Lecture 12: Smooth coverings, embedded manifolds

Wed 01 Oct 2025 12:21

Theorem 1.2.35. *Let $\pi : M \rightarrow N$ be a smooth submersion. Then any $F : N \rightarrow P$ is smooth if and only if $F \circ \pi$ is smooth. This is precisely universality of quotients in Man .*

Theorem 1.2.36. *Let $\pi : M \rightarrow N$ be a smooth submersion. If $F : M \rightarrow P$ is smooth and constant on each fiber $\pi^{-1}(n)$ of π , then there is exactly one $\tilde{F} : N \rightarrow P$ such that*

$$\begin{array}{ccc} M & \xrightarrow{F} & P \\ \pi \downarrow & \nearrow \tilde{F} & \\ N & & \end{array}$$

commutes.

Proof. Exists by topology, smooth by theorem 1.2.36. ■

Theorem 1.2.37. *Let there be a diagram of solid arrows*

$$\begin{array}{ccc} & M & \\ \pi_1 \swarrow & & \searrow \pi_2 \\ N_1 & \xrightarrow{\quad F \quad} & N_2 \end{array}$$

where each solid arrow is a submersion which are constant on each others fibers. Then there is a unique diffeomorphism as indicated by the dashed arrow, rendering the diagram commutative.

Definition 1.2.38. Recall that a covering space of a space M is a surjective, continuous map $\pi : E \rightarrow M$ between connected and locally path-connected spaces, such that for all $p \in M$, there is a neighborhood U of p such that each component of $\pi^{-1}(U)$ is mapped homeomorphically to U . We say U is *evenly covered*. If E is also simply connected (all loops contract to a point), we say it is a *universal cover*.

Proposition 1.2.39. *Let $\pi : E \rightarrow M$ be a smooth covering map. Given any evenly covered open $U \subseteq M$, any $q \in U$, and any $p \in \pi^{-1}(q)$, then there is a unique local section $\sigma : U \rightarrow E$ such that $\sigma(q) = p$.*

Proposition 1.2.40. *Let M be a smooth manifold and $\pi : E \rightarrow M$ a (not necessarily smooth) covering. Then E is a topological n -manifold with a unique smooth structure promoting π to a smooth covering.*

Corollary 1.2.41. *If M is a connected smooth manifold, then there is a simply connected smooth manifold $\tilde{M}$ together with a smooth map $\pi : \tilde{M} \rightarrow M$. We call $\tilde{M}$ the universal smooth cover.*

Chapter 2

Submanifolds

2.1 Embedded submanifolds

Definition 2.1.1. Let M be a smooth manifold, and $S \subseteq U$ a subspace endowed with a smooth structure such that the inclusion $S \hookrightarrow U$ is a smooth embedding. Then S is an *embedded submanifold*.

Lecture 13: Submanifolds

Fri 03 Oct 2025 12:20

For example, $S^n \hookrightarrow \mathbb{R}^n$ is an embedded submanifold. The main proposition is

Proposition 2.1.2. *Let $F : N \rightarrow M$ be a smooth embedding. Then $\text{im } F$ is an embedded submanifold of M with a unique smooth structure.*

Proof. Endow $\text{im } F$ with the smooth structure on N postcomposed by F or something. ■

Let $U \subseteq M$ and $f : U \rightarrow M$ smooth. Then $(1, f) : U \rightarrow M \times N$ is a homeomorphism onto the image $f \subseteq M \times N$. This is a section of the projection $\pi : M \times N \rightarrow M$, so $(1, f)$ is an immersion. We then have that the graph is an embedded submanifold. We will see that embedded submanifolds look locally like graphs of functions later.

Definition 2.1.3. An embedded submanifold $S \subseteq M$ is *properly embedded* if and only if $S \hookrightarrow M$ is a proper map (preimage of compact sets is compact).

Proposition 2.1.4. *Let $S \hookrightarrow M$ be an embedded submanifold. Then S is a proper embedding if and only if S is closed in M .*

Proof. If $S \hookrightarrow M$ is properly embedded, then the inclusion is closed since manifolds are locally compact Hausdorff (Lee A.57). Conversely, Lee A.53. ■

In particular, every compact/closed embedded submanifold is a proper embedding. In particular,

Proposition 2.1.5. *Let $F : M \rightarrow N$ be smooth. Then the graph is properly embedded in $M \times N$.*

Proven in the homework last year that graphs of continuous maps are closed.

Definition 2.1.6. A k -dimensional slice S of $U \subseteq \mathbb{R}^n$ is of the form $\{(x^\mu, c^\nu)\}$ for the c^μ 's constant.

Definition 2.1.7. Let M be a smooth manifold with a chart (U, φ) . If $S \subseteq U$ such that $\varphi(S)$ is a k -slice of $\varphi(U)$, then S is a k -slice of U .

Theorem 2.1.8 (Slice charts). *All embedded submanifolds locally look like $\mathbb{R}^k \hookrightarrow \mathbb{R}^{n \geq k}$. In particular, if $S \hookrightarrow U$, then for all $p \in S$ there is a chart (U, φ) such that $S \cap U$ is a k -slice of U .*

proof is like rank thm looks like slices.

Definition 2.1.9. Let $\Phi : M \rightarrow N$ be smooth. Then for any $c \in N$, the fiber $\Phi^{-1}(c)$ is the *level set* of Φ at c .

Proposition 2.1.10. *Let $\Phi : M \rightarrow N$ be smooth with constant rank. Then the level sets of Φ are properly embedded submanifolds in M .*

Lecture 14: More submanifolds

Mon 06 Oct 2025 12:21

Recall the main theorem. The slice charts theorem tells us that an embedded submanifold has nice charts which look like $(x^1, \dots, x^k) \mapsto (x^1, \dots, x^k, 0, \dots, 0)$.

Definition 2.1.11. Let $p \in M$ and $\Phi : M \rightarrow N$ smooth. p is a *regular point* of Φ if $d_p\Phi : T_pM \rightarrow T_{\Phi(p)}N$ is surjective. Otherwise, p is a *critical point* of Φ . A $c \in N$ is a *regular value* of Φ if p is a regular point for all $p \in \Phi^{-1}(c)$. If c is a regular value of Φ , then the fiber of c is a *regular level set* of Φ .

Corollary 2.1.12. *Every regular level set between smooth manifolds is a proper embedded submanifold of codimension equal to the dimension of the domain.*

Lecture 15: Lie groups

Wed 08 Oct 2025 12:20

We conclude submanifolds by discussing their tangent spaces. Let $S \hookrightarrow M$ be an immersed submanifold. The inclusion i raises to an injection $d_p(i) : T_pS \rightarrow T_pM$ for each $p \in S$. Since it's injective, we may identify it with its image up to isomorphism. This idea yields a useful proposition.

Proposition 2.1.13. *Let $p \in S \hookrightarrow M$ be an embedded submanifold. Then T_pS is isomorphic to the subset K of T_pM such that for each $v \in K$ and $f \in C^\infty(M)$ which vanishes on S ($f|_S = 0$), we have $v(f) = 0$.*

Proof. Let $v \in T_pS$ identified with a subspace of T_pM . Then there is exactly one $w \in T_pS$ with $v = d_p(i)(w)$. If $f|_S = 0$, then $v(f) = w(f \circ i) = w(0)$. For the converse, let $v \in T_pM$ satisfying the criteria. Choose slice coordinates $(x^1, \dots, x^n)$ for S in a neighborhood U of p . Then $i : S \cap U \rightarrow M$ has coordinate representation $\hat{i}(x^1, \dots, x^k) = (x^1, \dots, x^k, 0, \dots, 0)$. Then

$$v = v^\mu \partial_\mu|_p.$$

We have $v^j = 0$ whenever $j > k$. Choose some $f(x)$ with $f|_{U \cap S} = 0$. It just kind of follows. ■

Corollary 2.1.14. *Let $S \hookrightarrow M$ be an embedded submanifold, $U \subseteq M$, and $\Phi : U \rightarrow N$ any local defining map for S (meaning $\Phi^{-1}(c) = S \cap U$ for some c). Then $T_pS = \text{Ker } d_p\Phi$.*

This is quite useful. Suppose that $S \subseteq M$ is a level set of some smooth submersion $\Phi = \Phi^\mu : M \rightarrow \mathbb{R}^k$. Then any $v \in T_pM$ is also tangent to S if $v(\Phi^i) = 0$ for all i .

Chapter 3

Lie groups

Definition 3.0.1. A *Lie group* is a topological group G with a smooth structure such that multiplication and inversion are smooth. Equivalently, G is a group object in the category of smooth manifolds.

Proposition 3.0.2. Let G be a smooth manifold together with a smooth map $G \times G \rightarrow G$ given by $(g, h) \mapsto gh^{-1}$. Then G is a Lie group, and multiplication and inversion are recovered in the obvious way.

Definition 3.0.3. A *left translation* by $g \in G$ is the map $L_g : G \rightarrow G$ given by $h \mapsto gh$. Right translations similarly are $R_g : h \mapsto hg$. They are diffeomorphisms since $L_g^{-1} = L_{g^{-1}}$.

A useful example is $\mathrm{GL}(n, \mathbb{R})$ or $\mathrm{GL}(n, \mathbb{C})$. These are of dimension n^2 and $2n^2$, respectively. Most useful Lie groups arise as subgroups of these. We can also consider $\mathbb{R}^n, \mathbb{C}^n$ under addition or $\mathbb{R}^*, \mathbb{C}^*$ under multiplication. The latter have Lie subgroups $R_{>0}$ and S^1 .

Proposition 3.0.4. Let G, H be Lie groups. Then $G \times H$ is a Lie group.

Definition 3.0.5. A *discrete* Lie group is a 0-dimensional Lie group.

Definition 3.0.6. Lie groups form a category $\mathcal{Lie}$. A morphism $F : G \rightarrow H$ of Lie groups is a smooth group homomorphism.

Exponentiation $(x^1, \dots, x^n) \mapsto (e^{x^1}, \dots, e^{x^n})$ is a Lie group homomorphism $\mathbb{R}^n$ to the torus $\mathbb{T}^n$. Conjugation is smooth and is thus a Lie group homomorphism.

Theorem 3.0.7. Lie group homomorphisms have constant rank.

Proof. Choose $g \in G$ a Lie group with identity 1. Since $F : G \rightarrow H$ is a Lie group hom, $F(1) = 1$. We will show the rank of $d_g F$ is $d_1 F$, and this suffices. Choose $h \in G$, and consider $(F \circ L_g)(h) = F(gh) = (L_{F(g)} \circ F)(h)$. Thus,

$$F \circ L_g = L_{F(g)} \circ F.$$

We can pass to the differential

$$d_1(F) \circ d_1 L_g = d_1(L_{F(g)}) \circ d_1(F).$$

Since each L is a diffeomorphism, we see that the differentials of L are constant rank. Apparently it just follows idk check the book. ■

Corollary 3.0.8. A Lie group homomorphism is an isomorphism if and only if it is bijective.

This follows by the global rank theorem.

Remark 3.0.9. The kernel $\text{Ker } F$ of a Lie group homomorphism is the *level set* $F^{-1}(1)$. The fact that F is constant rank tells us that the kernel is a *properly embedded submanifold* of the domain. Since the kernel is also a group, it should be a Lie group.

Lecture 16: More Lie groups

Fri 10 Oct 2025 12:22

Definition 3.0.10. A Lie subgroup is a subgroup with a smooth structure and topology such that it is a Lie group, and inclusion is an immersed submanifold.

Proposition 3.0.11. *Let G be a Lie group and H a subgroup that is an embedded submanifold.*

Proof. It suffices to show H is a Lie group. This follows easily by a diagram chase. ■

Lemma 3.0.12. *Let G be a Lie group and H an open subgroup. Then H is an embedded Lie group, and H is closed, and H is a union of connected components.*

Proof. We get that H is an embedded submanifold of codimension 0, and we appeal to proposition 3.0.11. The coset gH is the image of left multiplication L_g , which is a diffeomorphism, so the image of the open set H is open. We of course have $G \setminus H$ is the union of all nontrivial cosets, so H is the complement of the open set $G \setminus H$. The final follows from usual topology: any clopen set is a union of path components. Of course this coincides with path connectedness in manifolds. ■

For example, given an open subset U of S^1 containing 1, the group $\langle U \rangle$ is an open subgroup of G .

Proposition 3.0.13. *Let G be a Lie group, and $W \subseteq G$ a neighborhood of 1. Then W generates an open subgroup of G . If W is connected, then W generates a connected open subset of G . If G is connected, then W generates G .*

Proof. Let $H = \langle W \rangle$. Define

$$W_k = \left\{ g \in G \mid \exists \{w_i\}_{i=1}^\ell \subseteq W, \ell \leq k \text{ s.t. } g = \prod_{i=1}^\ell w_i \right\}.$$

Then

$$H = \bigcup_{k \geq 1} W_k.$$

Since inversion is a diffeomorphism, $W^{-1} = i(W)$ is open. Of course $W_1 = W \cup W^{-1}$, and $W_k = W_{k-1}W_1$. This is just

$$W_k = W_{k-1}W_1 = \bigcup_{g \in W_1} L_g(W_{k-1}).$$

Since L_g is a diffeomorphism, each of these sets is open, so W_k is open. Subgroup follows by $1 \in W$. H is then a Lie subgroup by lemma 3.0.12. If W is connected, then so is W^{-1} , and thus $W_1 = W \cup W^{-1}$ is connected since $1 \in W \cap W^{-1}$. We then see

$$W_2 = \mu(W_1 \times W_1)$$

is connected since μ is continuous. Inducting gives W_k is connected. Since each of these sets contains 1, they are not disjoint, so H is connected. If G is connected, then H is an open and closed subgroup so $H = G$. ■

Definition 3.0.14. Let G be a Lie group, and let $G_0 \subseteq G$ be the connected component containing 1. We call G_0 the *identity component* of G .

Proposition 3.0.15. Let G be a Lie group and G_0 the identity component. Then G_0 is normal in G , and is the only connected subgroup. Every connected component of G is diffeomorphic to G_0 .

The idea of the proof is that connected components are cosets. Act by L_g and it falls out.

Proposition 3.0.16. If $F : G \rightarrow H$ is a Lie group homomorphism, then $\text{Ker } F$ is a properly embedded Lie subgroup of codimension $\text{rk } F$.

Proof. We get subgroup from group theory. The kernel is the fiber of the identity. Since F has constant rank, the kernel is a properly embedded submanifold. ■

Proposition 3.0.17. Let $F : G \rightarrow H$ be an injective Lie group homomorphism. Then $F(G)$ has a unique smooth structure such that it is a Lie subgroup of H . Furthermore, $G \cong F(G)$.

Proof. Because F has constant rank, it is (easily shown to be) an injective smooth submersion. Therefore, its image admits a unique smooth structure making it an immersed submanifold, and thus a Lie subgroup. The rest follows. ■

For example, $\text{SL}(n, \mathbb{R})$ is the kernel of $\det : \text{GL}(n, \mathbb{R}) \rightarrow \mathbb{R}$.

Theorem 3.0.18. Let G be a Lie subgroup with $H \subseteq G$ a Lie subgroup. Then H is closed in G if and only if it is embedded.

It will be useful to consider *smooth* group actions of Lie groups on smooth manifolds. The best formalism of this is as a smooth map $G \times M \rightarrow M$ which induces the relevant group homomorphism.

Consider $O(n)$ as the subset of $\text{GL}(n, \mathbb{R})$ such that $\langle Ax, Ay \rangle = \langle x, y \rangle$. We call this the orthogonal group. We will define this somehow eventually.

Lecture 17: More more Lie groups

Tue 14 Oct 2025 12:20

Suppose we have $H \subseteq G$ a Lie subgroup. Then H acts smoothly on G by right, multiplication or left multiplication. The orbits under this action are cosets. We call the stabilizer of $p \in M$ the *isotropy* (sub)group at p .

Definition 3.0.19. Let M, N be manifolds with G -actions $\theta : G \times M \rightarrow M$, $\psi : G \times N \rightarrow N$ for G a Lie group. A smooth map $F : M \rightarrow N$ is called *G -equivariant* if the diagram

$$\begin{array}{ccc} M & \xrightarrow{F} & N \\ \theta_g \downarrow & & \downarrow \psi_g \\ M & \xrightarrow{F} & N \end{array}$$

commutes for each $g \in G$. The (functor) category Smth^G is the category of smooth G -manifolds together with G -equivariant maps.

Theorem 3.0.20 (Equivariant rank theorem). Let $F : M \rightarrow N$ be a smooth G -equivariant map for M, N G -manifolds, G a Lie group. If the action on N is transitive, then F is constant rank, the action on M is transitive, and if F is (sur/in/bi)jective then it is (sub/im)mersion/diffeomorphism.

Proposition 3.0.21. Given a G -action θ on M , for all $p \in M$ the orbit map $\theta^{(p)} : G \rightarrow M$ given by $g \mapsto gp$ is smooth and has constant rank. Since fibers are isotropy groups, they are properly embedded.

Lecture 18: Vector fields

Wed 15 Oct 2025 12:44

Recall that given a manifold M , a smooth vector field is a section $V : M \rightarrow TM$ of the projection of the tangent bundle $\pi : TM \rightarrow M$. The field V at a point $p \in M$ then takes the form

$$V_p = V_p^i \partial_i|_p.$$

We can then define vector field in ways like

$$x \mapsto x^i \partial_i|_x.$$

If we choose a different chart on M , we could obtain a different coordinate function ∂_i and get new fields.

Lecture 19: idk

Fri 17 Oct 2025 12:20

Let $F : M \rightarrow N$ be a smooth map. For each $p \in M$, we get an induced map of tangent spaces $dF_p : T_p M \rightarrow T_{F(p)} N$. However, if V is a smooth vector field on M , we cannot in general define dFV as a smooth vector field on N .

Definition 3.0.22. Let $F : M \rightarrow N$ be smooth, and let $X : M \rightarrow TM$ and $Y : N \rightarrow TN$ be vector fields. Then X and Y are F -related if and only if for all $p \in M$, $dF_p(V_p) = Y_p$, where of course $X_p = X|_{T_p M}$, and the same for Y .

This tells us that $X(f \circ F) = Y(f) \circ F$.

Proposition 3.0.23. Let $F : M \rightarrow N$ be a diffeomorphism. Then for all vector fields $X : M \rightarrow TM$, there exists a unique vector field $Y : N \rightarrow TN$ which is F -related to X .

Definition 3.0.24. Let X, Y be smooth vector fields on M . Define the Lie bracket $[X, Y] := (X \circ Y) - (Y \circ X)$.

Definition 3.0.25. A Lie algebra is a $(\mathbb{R})$ -vector space $\mathfrak{g}$ together with a Lie bracket $[-, -] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ which is antisymmetric, bilinear, and satisfies the Jacobi identity. A morphism of Lie algebras is a linear map $f : \mathfrak{g} \rightarrow \mathfrak{h}$ such that $f[X, Y]_{\mathfrak{g}} = [f(X), f(Y)]_{\mathfrak{h}}$.

To any manifold M is associated a Lie algebra $\mathfrak{X}$ of smooth vector fields, with Lie bracket given by definition 3.0.24. Moreover, the Lie bracket is natural in the following sense: suppose $F : M \rightarrow N$ is smooth, and suppose $X_1, X_2 \in \mathfrak{X}(M)$, $Y_1, Y_2 \in \mathfrak{X}(N)$ are smooth vector fields with X_i F -related to Y_i . Then the Lie bracket $[X_1, X_2]$ is also F -related to $[Y_1, Y_2]$. In particular, diffeomorphisms $F : M \cong N$ induce Lie algebra homeomorphisms. A corollary of this is if $S \subseteq M$ is immersed, then the Lie bracket of two vector fields tangent to S is also tangent to S .

Proposition 3.0.26. Given a Lie group G , there exists a natural finite dimensional Lie algebra $\mathfrak{g}$ such that $\mathfrak{g} \subseteq \mathfrak{X}(G)$ and $[\mathfrak{g}, \mathfrak{g}] \subseteq \mathfrak{g}$.

Definition 3.0.27. Given a Lie group G and a vector field $V \in \mathfrak{X}(G)$, we say V is a *left-invariant* vector field on G if and only if $L_{g*} V = V$ for all $g \in G$. Right invariance is defined similarly. The Lie algebra of proposition 3.0.26 is given by the set of left-invariant vector fields.

Lecture 20: The Lie algebra of a Lie group

Mon 20 Oct 2025 12:21